PHASE 1

Screenshot 1A — object tests

```
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe
9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

Screenshot 1B — object storage structure

```
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$
```

PHASE 2

Screenshot 2A — tree tests

```
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$
```

Screenshot 2B — raw tree object

```
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ find .pes/objects -type f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/3d/8c0574116651ef38e98c4feb322ca752c22ef27746734cf24242a7625d708b
.pes/objects/ea/3b6ce0df88dc227083adba05c43d15da3a824a98da9bcaaf1930277fb07f30
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ xxd .pes/objects/2c/f8d83d9ee2
9543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 | head -20
00000000: 626c 6f62 2036 0068 656c 6c6f 0a         blob 6.hello.
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$
```

PHASE 3

Screenshot 3A — init + add + status

```
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ echo "hello" > file1.txt
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ echo "world" > file2.txt
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes add file1.txt file2.txt
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt
```

Screenshot 3B — index file

```
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776359869 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776359883 6 file2.txt
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$
```

PHASE 4

Screenshot 4A — commit log

```
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ echo "Hello" > hello.txt
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes add hello.txt
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes commit -m "Initial commit"
Committed: 9dc1f29e530d... Initial commit
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ echo "World" >> hello.txt
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes add hello.txt
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes commit -m "Add world"
Committed: ac08046bab68... Add world
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ echo "Bye" > bye.txt
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes add bye.txt
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes commit -m "Add farewell"
Committed: e473328463ef... Add farewell
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ ./pes log
commit e473328463effb0ed14112b674d6db7e1f1cdbad4eae22f66df0b2fb70b3abff
Author: PES User <pes@localhost>
Date:   1776360208

    Add farewell

commit ac08046bab68acdd49df7120556594bb1118405ff40c1dc124457cea22c2ef12
Author: PES User <pes@localhost>
Date:   1776360163

    Add worldell

commit 9dc1f29e530db51a352db685771109270dccd2e9397c1a0e886dd4e43de375b4
Author: PES User <pes@localhost>
Date:   1776360106

    Initial committial commiQ
```

Screenshot 4B — object growth

```
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/9d/c1f29e530db51a352db685771109270dccd2e9397c1a0e886dd4e43de375b4
.pes/objects/ac/08046bab68acdd49df7120556594bb1118405ff40c1dc124457cea22c2ef12
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e4/73328463effb0ed14112b674d6db7e1f1cdbad4eae22f66df0b2fb70b3abff
.pes/objects/f3/97707e14d1cabf1c29a72d833bffb94d4a474ecc48789d6496742ae6541034
.pes/refs/heads/main
```

Screenshot 4C — HEAD + branch

```
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ cat .pes/refs/heads/main
e473328463effb0ed14112b674d6db7e1f1cdbad4eae22f66df0b2fb70b3abff
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

FINAL — integration test

```
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 4a479d067d83... Initial commit

Log after first commit:
commit 4a479d067d83cd006c5449d04fe2fc3032249571a884b88a39df80ddf7515c59
Author: PES User <pes@localhost>
Date:   1776360352

    Initial commit


--- Second Commit ---
```

```
--- Second Commit ---
Committed: 3da594b1467d... Update file.txt

--- Third Commit ---
Committed: e6fe41850faf... Add farewell

--- Full History ---
commit e6fe41850fafb3b232d34694d422014571afd81dc1cf8b7e9adc0f266b994c91
Author: PES User <pes@localhost>
Date:   1776360352

    Add farewell

commit 3da594b1467de084135b29204cc17103640205007e84582e026547a4f9e66d55
Author: PES User <pes@localhost>
Date:   1776360352

    Update file.txt

commit 4a479d067d83cd006c5449d04fe2fc3032249571a884b88a39df80ddf7515c59
Author: PES User <pes@localhost>
Date:   1776360352

    Initial committial commiQ
```

```
--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
e6fe41850fafb3b232d34694d422014571afd81dc1cf8b7e9adc0f266b994c91

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/3d/a594b1467de084135b29204cc17103640205007e84582e026547a4f9e66d55
.pes/objects/4a/479d067d83cd006c5449d04fe2fc3032249571a884b88a39df80ddf7515c59
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e6/fe41850fafb3b232d34694d422014571afd81dc1cf8b7e9adc0f266b994c91

=== All integration tests completed ===
sufiya@sufiya-VirtualBox:~/PES1UG24CS919-pes-vcs$
```

Phase 5

Q5.1

To implement pes checkout <branch>, we update .pes/HEAD to point to the selected branch and read its commit hash. Then we load the commit's tree and update the working directory to match it. The index is also updated accordingly.

This is complex because we must handle file additions/deletions and avoid overwriting uncommitted changes.

Q5.2

To detect a dirty working directory, we compare each file's current hash (from the working directory) with the hash stored in the index. If they differ, the file has uncommitted changes. If such a file also differs in the target branch, checkout should be refused.

Q5.3

In a detached HEAD state, HEAD points directly to a commit instead of a branch. New commits can still be made, but they are not referenced by any branch.
These commits can be recovered by creating a new branch pointing to them.

Phase 6

Q6.1

To find unreachable objects, we start from all branch heads and traverse commits, trees, and blobs, marking each visited object.
A hash set can be used to track reachable objects efficiently. Any object not visited can be deleted.
This may involve visiting hundreds of thousands of objects in large repositories.

Q6.2

Running garbage collection during a commit can cause a race condition. If a new object is created but not yet linked to a branch, GC may delete it as unreachable.
Git avoids this using locking and by ensuring GC runs only when it is safe.